

Tutor quality is not tutor-student fit: Evidence from online secondary-school tutoring

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Abstract

Online tutoring platforms recruit and screen tutors on quality, yet families often explain a failed tutoring relationship by saying the tutor did not suit the student. This study asks whether perceived tutor quality and perceived tutor–student fit are empirically distinct attributes. Administrative records were obtained for 220 tutoring relationships on one online International General Certificate of Secondary Education tutoring platform; student questionnaires were returned for 178 of these, parent questionnaires for 163, and both for 134 matched student–parent pairs. Quality and fit were essentially uncorrelated ($r = .12$), loaded on separate factors, and both contributed independently to student satisfaction (fit $\beta = .51$; quality $\beta = .18$). Their variance structures differed: perceived quality showed substantial clustering by tutor (intraclass correlation .31) whereas perceived fit showed essentially none ($< .01$), consistent with quality having a tutor-level component and fit varying primarily within a tutor's own caseload. On the parent side, perceived value and platform responsiveness were each independently associated with parent satisfaction, whereas the session fee added no independent association. Findings are associational and from a single platform. They imply that screening tutors on quality cannot substitute for matching, because the two attributes do not vary at the same level.

Keywords: online tutoring; tutor–student fit; tutor quality; student satisfaction; shadow education

1. Introduction

Private supplementary tutoring, often called shadow education because it tracks the curriculum of formal schooling, is established across many education systems (Luo & Chan, 2022; Zhang & Bray, 2020), and digital delivery has widened its reach further. A substantial literature asks whether tutoring raises attainment (Nickow et al., 2020; Zhang & Liu, 2022), including on remote delivery (Carlana & La Ferrara, 2025; Gortazar et al.,

2024). Much less is known about what makes a tutoring relationship work once it has begun — the question a platform must solve each time it assigns a tutor.

Platforms solve that problem almost entirely through screening: tutors are recruited, tested, rated and ranked, and the resulting quality signal decides who is offered to whom. The implicit assumption is that a better tutor produces a better tutoring relationship. Families do not always describe it that way. A recurring complaint in platform records is that the tutor was knowledgeable but did not suit the student — the pace was wrong, the explanations did not land. If that account is correct, tutor quality and tutor–student fit are two different things, and a platform that measures only the first is missing information it needs.

The distinction is familiar in principle. Matching research separates vertical quality, on which everyone agrees more is better, from horizontal fit, which is specific to a pairing and on which people disagree (Du & Lei, 2022). Online-learning research on satisfaction, however, has tended to fold both into composite instructor-quality scales (Al-Fraihat et al., 2020; Khan et al., 2023; Martin & Bolliger, 2022), so the two are rarely measured separately in the same sample and their relationship is rarely tested. Whether they are empirically separable in a real tutoring population, and whether they behave differently, is an open question.

This study addresses it using administrative records and separate student and parent questionnaires from one online International General Certificate of Secondary Education (IGCSE) tutoring platform. The primary contribution is evidence that perceived quality and perceived fit are distinct constructs that are distributed differently: they are close to uncorrelated, load on separate factors, are each independently associated with student satisfaction, and — the point with the clearest operational consequence — do not cluster in the same way. Perceived quality varies substantially between tutors, as a stable attribute of the tutor should; perceived fit does not, varying instead within a tutor's own caseload, as an attribute of the pairing should.

Two further sets of findings are reported at lower confidence. The parent questionnaire allows the same separation test on the payer's side, distinguishing perceived value from platform responsiveness and both from the price paid; and because the platform records whether a further paid booking followed, evaluations can be linked to behaviour rather than to stated intention alone. Whether the student's or the parent's evaluation carries more information about that outcome is treated as exploratory, for reasons set out in Section 3.7.

2. Conceptual background and hypotheses

2.1 Quality and fit as different attributes

The matching literature distinguishes vertical from horizontal attributes. A vertical attribute is one on which preferences are aligned: every student would prefer the tutor who explains more clearly, prepares more thoroughly and gives more useful feedback. A horizontal attribute is one on which preferences differ: the pace that suits one student is too slow for a second and too fast for a third, and no ordering of tutors captures this. Du and Lei (2022) formalise the distinction for matching platforms and show that the two require different information; a platform can rank on the vertical dimension but must pair on the horizontal one.

Educational research supplies content on both sides. The tutoring literature attributes effectiveness to instructional behaviours that are plausibly stable across students: clarity of explanation, preparation, formative feedback and responsiveness to error (Kraft & Falken, 2021; Nickow et al., 2020), and work on one-to-one online tutoring shows tutors varying systematically in these behaviours across their caseloads (Wang et al., 2025a, 2025b). This is tutor quality as a vertical attribute. Studies of instructional matching instead emphasise pace, prior-knowledge alignment and interactional style, all properties of a specific pairing. This is tutor–student fit as a horizontal attribute.

Satisfaction research in online learning has largely not kept the two apart. Instruments derived from e-learning success and service-quality traditions typically combine items on clarity, preparation, pacing and suitability into a single instructor-quality construct (Al-Fraihat et al., 2020; Khan et al., 2023), and reviews of learner satisfaction treat instructor characteristics as one block (Lysitsa & Mavroeidis, 2024; Martin & Bolliger, 2022). Composites of this kind show high reliability whether the underlying attributes are one construct or two, so their success is not evidence that the distinction is unnecessary.

Two predictions follow. If quality and fit are distinct attributes rather than two labels for one evaluation, they should be weakly related and separable in the item structure, and each should carry information about satisfaction the other does not.

H1. Perceived tutor quality and perceived tutor–student fit are empirically distinct: weakly correlated and separable in the item structure.

H2. Perceived fit and perceived quality each contribute independently to student satisfaction.

2.2 Where the two attributes should vary

Distinctness in a correlation matrix is a weak test, since two constructs can be uncorrelated for uninteresting reasons. A stronger test uses the structure implied by the definitions. If quality is a property of the tutor, students taught by the same tutor should agree about it, so a meaningful share of the variance in perceived quality should lie between tutors. If fit is a property of the pairing, the same tutor should suit some students and not others, so almost none of the variance in perceived fit should lie between tutors. Where each tutor teaches several students these shares are directly estimable as intraclass correlations, and they discriminate the two accounts in a way a correlation cannot.

H3. A substantial share of the variance in perceived quality lies between tutors, whereas the variance in perceived fit is almost entirely within tutors.

2.3 The payer's side: value, responsiveness and price

Supplementary tutoring is bought by parents and consumed by students, so it is evaluated by two people who see different parts of it (Al-Thani et al., 2024; Gupta, 2022a); buying-centre models describe exactly this separation of user and purchaser (Webster & Wind, 1972). The parent typically does not observe the lesson, but observes the price, the administrative service around it, and the child's reaction.

That suggests a parallel separation on the parent side, between perceived value and the responsiveness of the platform. Value in the sense of Zeithaml (1988) is a judgement about what was received relative to what was given up, and it is not the same as the price paid: two families paying identical fees can reach opposite conclusions about worth. Responsiveness is a service-delivery attribute in the SERVQUAL tradition (Parasuraman et al., 1988), concerning how quickly and effectively problems are handled, and evidence from Indian education providers indicates that parents attend to it closely (Narang & Bedi, 2025). If the parallel holds, value and responsiveness should be weakly related, both should be independently associated with parent satisfaction, and the fee paid should add nothing once value is accounted for.

H4. Perceived value and platform responsiveness each contribute independently to parent satisfaction, whereas the session fee does not.

2.4 Evaluation, operational events and continued participation

Expectation-disconfirmation accounts hold that satisfaction follows from the comparison of experience with expectation and is in turn associated with continuation (Bhattacharjee, 2001; Oliver, 1980). In tutoring, continuation is observable: the platform records whether a further paid booking took place. This permits a behavioural test rather than a test on

stated intention, which is the usual outcome in the satisfaction literature and which shares its measurement source with the predictors.

Two operational events are also observable. Early tutor reassignment is what a platform does when a pairing is reported not to be working, so if fit matters it should mark relationships at risk. The platform's initial response time to the enquiry, recorded in hours, is often treated as a service-quality lever, but among families who have already completed a paid lesson the enquiry is long past, so it is not obvious that it should carry information about continuation.

H5. Higher satisfaction is associated with a further paid booking within 30 days.

H6. Early tutor reassignment is associated with lower continuation.

RQ1 (exploratory). Is objective initial response time associated with continuation among relationships in which a first lesson was completed? This is posed as a question rather than as a hypothesis of no effect, because a non-significant coefficient cannot by itself establish the absence of an association.

2.5 An exploratory question

Because student and parent satisfaction were measured from different respondents and can be entered into the same model, it is possible to ask which carries more unique information about continuation. The question is worth asking, since the shadow-education literature emphasises the parent as decision-maker while the online-learning literature places the learner at the centre. It is reported as exploratory rather than hypothesis-testing for two reasons given in Section 3.7: limited power to distinguish two correlated predictors in 134 matched cases, and the parent satisfaction scale being the least reliable measure in the study.

3. Method

3.1 Design and setting

The study uses a retrospective quantitative design combining the administrative records of a single online IGCSE tutoring platform with separate student and parent questionnaires, administered between 10 May 2026 and 10 June 2026 and linked to those records by pseudonymous identifier. Questionnaires were completed within 15 days of each student's first completed lesson, so perceptions were measured early in the relationship; the analytical file holds no exact completion dates, so strict temporal ordering with the 30-day booking outcome cannot be established and a repeat booking

could in principle have preceded questionnaire completion. Associations between questionnaire measures and booking outcomes are therefore described throughout as concurrent associations rather than as predictors, and are not interpreted causally.

3.2 Population, eligibility and sample

The population is students receiving IGCSE-level tutoring through the platform. The tutor pool combines peer tutors, typically students who have recently taken the relevant qualification, with adult or professional tutors; tutor type was not recorded and could not be analysed separately. A relationship was eligible if the student had completed at least one paid session in an IGCSE-level academic subject, enough time had elapsed to observe the outcome, and records identified the first session and later bookings. Relationships were excluded where the first lesson was never completed or the arrangement was a single-session service from the outset. These rules were fixed before outcomes were analysed.

The dataset contains 220 tutoring relationships involving 220 unique students aged 14 to 17 ($M = 15.46$, $SD = 0.83$), 94 in Year 10 and 126 in Year 11; each student contributes only one relationship to the analytical dataset, so no student appears more than once and there is no student-level clustering above the relationship. Mathematics was the most common subject ($n = 72$), followed by Physics (37), Chemistry (35), Biology (30), English (28) and Economics (18). Student questionnaires were returned for 178 relationships (80.9%), parent questionnaires for 163 (74.1%) and both for 134 (60.9%). The unit of analysis is the tutoring relationship, and because each student contributes exactly one, it is equivalently the student. The 220 relationships involve 24 initial tutors, each teaching four to 17 students in the sample; this structure makes the variance decomposition in H3 estimable. Twenty-nine relationships (13.2%) involved a tutor change within 30 days. Median initial response time was 4.15 hours (range 0.7 to 38.0) and mean time from enquiry to first lesson was 4.03 days ($SD = 1.95$). Table 1 summarises the sample.

Table 1. Sample characteristics, overall and by 30-day repeat-booking status

Characteristic	All ($n = 220$)	Repeat booking ($n = 127$)	No repeat booking ($n = 93$)	p
Student age, M (SD)	15.46 (0.83)	15.35 (0.82)	15.62 (0.82)	.014
Year 11, n (%)	126 (57.3)	67 (52.8)	59 (63.4)	.149
Subject: Mathematics, n (%)	72 (32.7)	41 (32.3)	31 (33.3)	.268 (a)

Subject: Physics, Chemistry or Biology, n (%)	102 (46.4)	63 (49.6)	39 (41.9)	
Subject: English or Economics, n (%)	46 (20.9)	23 (18.1)	23 (24.7)	
Trial session held, n (%)	76 (34.5)	46 (36.2)	30 (32.3)	.640
Tutor experience, months, M (SD)	21.9 (16.0)	21.4 (16.3)	22.6 (15.7)	.583
First-session fee, local units, M (SD)	19.42 (4.41)	18.87 (3.99)	20.16 (4.85)	.037
Initial response time, hours, Mdn (range)	4.15 (0.7 to 38.0)	3.60 (0.7 to 38.0)	4.50 (1.0 to 38.0)	.127 (b)
Days to first lesson, M (SD)	4.03 (1.95)	4.00 (1.96)	4.06 (1.95)	.809
Early tutor change, n (%)	29 (13.2)	9 (7.1)	20 (21.5)	.003
Student questionnaire returned, n (%)	178 (80.9)	109 (85.8)	69 (74.2)	.046
Parent questionnaire returned, n (%)	163 (74.1)	101 (79.5)	62 (66.7)	.046
Both questionnaires returned, n (%)	134 (60.9)	86 (67.7)	48 (51.6)	.023

Note. Repeat booking = at least one further paid booking within 30 days of the first completed lesson. *p* from Welch *t*-tests and chi-square tests; (a) chi-square on the six-category subject variable, (b) Mann-Whitney *U*. Fee is in local currency units. The final three rows constitute the non-response analysis described in Section 3.6.

3.3 Measures

All attitudinal items used a five-point agreement scale (1 = strongly disagree, 5 = strongly agree) and composite scores are item means. The student questionnaire measured tutor–student fit (four items), tutor quality (four items) and student satisfaction (two evaluative items); the parent questionnaire measured perceived value (three items), platform responsiveness (three items) and parent satisfaction (two evaluative items). Wording drew on the responsiveness dimension of SERVQUAL (Parasuraman et al., 1988), Zeithaml's (1988) conceptualisation of value, expectation-disconfirmation theory (Oliver, 1980) for satisfaction, and instructor-quality items in Al-Fraihat et al. (2020) and Khan et al. (2023) for quality; fit items were written for this study following Du and Lei's (2022) definition of horizontal fit. Because separating quality from fit is the object of the study, the two blocks keep the tutor's instructional conduct (clarity, preparation, feedback, adaptation)

apart from the suitability of that conduct for this student (style, pace, needs, overall match). Full item wording is in Appendix A.

Because 29 relationships (13.2%) involved a tutor change within 30 days and questionnaires were completed within 15 days of the first lesson, the referent of the tutor-directed items matters for the variance decomposition, which groups responses by the initial tutor. All student questionnaire items referred to the initial tutor recorded for the relationship, including where a reassignment had occurred before questionnaire completion: students who had already been reassigned were instructed to evaluate the original tutor and the original relationship rather than the replacement. Every rating is therefore attached to the same tutor used as the grouping factor, and the intraclass correlations reported in Section 4.1 are not affected by reassignment misclassification. A third item in each satisfaction block asks about intention to continue. Because these items overlap conceptually with the behavioural outcome, they are excluded from the primary satisfaction scales and analysed separately (Section 4.5).

3.4 Outcome variables

The primary behavioural outcome was 30-day repeat booking, coded 1 if at least one additional paid booking was completed within 30 days of the first completed lesson. Tutoring was normally arranged weekly, as single sessions or short blocks, so a 30-day window covers roughly four potential sessions and, for most families, at least one natural decision point. A secondary outcome, activity at 60 days, was coded 1 if at least one paid lesson fell in days 31 to 60; being later than any tutor change in the first 30 days, it is used where ordering matters.

3.5 Measurement quality and dimensionality

Following Cheung et al. (2024), measurement quality was assessed on several criteria rather than a single threshold. Cronbach's alpha was .747 for fit, .788 for quality, .819 for value and .750 for responsiveness. For the two-item satisfaction scales the Spearman–Brown coefficient is reported: .656 for student satisfaction ($r = .49$) and .583 for parent satisfaction ($r = .41$). Both indicate limited internal consistency, the parent measure markedly so, and this is the principal reason the student–parent comparison is treated as exploratory. Three-item versions including the intention items reach .765 and .707 but at the cost of criterion contamination, so they appear only as a sensitivity check.

Convergent and discriminant validity were assessed from three sources: the internal consistency of each scale, reported above; the factor structure and inter-factor correlations reported next; and, for the two focal student constructs, the variance decomposition across tutors reported in Section 4.1. Dimensionality was examined by

principal-axis factoring with promax rotation. For the 11 student items ($n = 178$), sampling adequacy was good ($KMO = .83$; Bartlett's $\chi^2(55) = 574.2$, $p < .001$) and three eigenvalues exceeded one (3.72, 2.12, 1.02; the fourth was 0.68). The three-factor solution reproduced the intended structure: the four fit items loaded .52 to .70, the four quality items .63 to .78 on a second factor, and the three satisfaction and intention items .60 to .79 on a third, with no cross-loading above .26 in absolute value; the two items closest in surface wording, FIT2 and TQ4, loaded cleanly (.70 and .63). The fit and quality factors correlated .15 (.19 restricted to the eight items); fit correlated .62 with the satisfaction factor and quality .36.

For the nine parent items ($n = 163$), sampling adequacy was also good ($KMO = .81$; Bartlett's $\chi^2(36) = 465.6$, $p < .001$). Value items loaded .77 to .82 and responsiveness items .62 to .79, each with cross-loadings below .17, and the two factors correlated .14 (.05 in a two-factor solution restricted to the six items). The satisfaction factor was weaker: PS1 loaded .87 and the intention item .54, but PS2 ("met the expectations I had") loaded only .35 with a cross-loading of $-.23$ on value, consistent with the low reliability of the parent satisfaction scale.

3.6 Non-response and common method variance

Because administrative records including the outcome are available for all 220 relationships, response status could be compared directly with booking behaviour. Responders were more likely to have booked again than non-responders: 61.2% versus 42.9% for the student questionnaire (continuity-corrected $\chi^2(1) = 3.98$, $p = .046$), 62.0% versus 45.6% for the parent questionnaire ($p = .046$) and 64.2% versus 47.7% for matched pairs ($p = .023$). In logistic models of response status, repeat booking was the only significant correlate of student response ($b = 0.77$, $p = .038$) and a marginal correlate of parent response ($b = 0.61$, $p = .070$); no other administrative variable was related to response. Data are therefore not missing completely at random, and the questionnaire-based models are estimated on a sample tilted towards families who booked again. The multiple-imputation analysis assumes missingness at random conditional on observed variables; that cannot be verified, but the variable most strongly associated with missingness is fully observed and is included as an auxiliary.

Predictors and satisfaction were reported by the same respondent, on the same instrument, at the same time. The satisfaction models (H2 and H4) are therefore same-source and cross-sectional, and common method variance may inflate them; this qualification applies to the fit and quality coefficients in Table 3. Three features limit but do not remove the risk: the repeat-booking outcome is behavioural and independently recorded, so H5, H6 and RQ1 do not share a source with the questionnaire predictors;

student and parent constructs come from different respondents; and predictor and criterion items refer to different objects. As a limited diagnostic, the first unrotated factor explained 33.8% of the variance in the student items and 37.3% in the parent items, which does not indicate one dominant factor but cannot rule out bias. The variance decomposition in H3 is not vulnerable in the same way, since it concerns how one construct is distributed across tutors rather than an association between two self-reported constructs.

3.7 Analytical strategy

Analysis proceeded in four stages. First, descriptives, reliabilities and correlations (Table 2). Second, ordinary least squares regression of student satisfaction on fit, quality, initial response time, trial-session status, year group and subject ($n = 178$), and of parent satisfaction on value, responsiveness, session fee, response time, year group and subject ($n = 163$). Third, linear random-intercept models with tutors as the grouping factor, estimating the share of variance in perceived quality and fit lying between tutors (H3). Fourth, binary logistic regression of 30-day repeat booking in four nested specifications, all on the 134 matched relationships so that changes in fit reflect added variables rather than a changing sample: an operational model with six administrative variables, that model plus student satisfaction, plus parent satisfaction, and plus both. Nested models were compared with likelihood-ratio tests and coefficients reported as odds ratios with 95% confidence intervals. Full-sample equivalents are in Appendix C.

Because 24 clusters is at the lower bound for cluster-robust inference, the primary tables report heteroskedasticity-consistent standard errors (HC3 linear, HC1 logistic), with tutor-clustered errors and random-intercept models as sensitivity analyses. Subject and year-group dummies are included in the satisfaction models but not in the primary booking models, because six further parameters on 86 events would reduce events per variable to about six. The primary matched model has 86 events and eight predictors, giving 10.8 events per variable, marginally above the conventional minimum; Riley et al. (2020) show this rule to be a coarse guide, a further reason the paper emphasises convergence across specifications rather than any single estimate.

Within each confirmatory family — the two satisfaction regressions, and the booking models — p-values for hypothesis-bearing terms are reported after Holm–Bonferroni adjustment as well as unadjusted. The student–parent comparison is not treated as confirmatory. With 134 matched cases, 86 events and two correlated satisfaction measures entered together, the design has limited power to distinguish their unique contributions, and the parent measure has a Spearman–Brown coefficient of .583, so any difference between them is partly a difference in measurement precision. That

comparison is therefore reported as exploratory throughout and is not used to support the paper's claims. Multicollinearity is not a concern in either satisfaction model. The two focal predictors in the student model correlate .12 and those in the parent model correlate .03, so each pair is close to orthogonal, and variance inflation factors computed across all terms in both models reached a maximum of 1.40 (student model 1.34; focal terms: fit 1.05, quality 1.08, value 1.19, responsiveness 1.40), well below any conventional threshold. Analyses used Python 3.12 with statsmodels 0.14.6.

3.8 Ethics, consent and researcher role

Ethical approval was granted by the Zigma Six Ethics Review Board, India, on 15 April 2026 (Approval No. SS/2026/002-23; OHRP IRB00015192, Federalwide Assurance FWA00036459). Administrative booking records were followed for 60 days after each relationship's first completed lesson, with the final observation on 9 August 2026. All participating students were under 18, so written parental or guardian consent was obtained together with student assent. Participation was voluntary and participants were told their decision would not affect their access to or relationship with the platform. Use of pseudonymised administrative records for all 220 relationships, including those without a questionnaire, was authorised under the same approval. Names, contact details and other directly identifying information were excluded from the analytical dataset, and records were linked by pseudonymous identifier.

The second author is the founder of the tutoring platform from which the data were obtained. This provided access to operational records and created a potential conflict of interest. To limit researcher bias, hypotheses, eligibility criteria, outcome definitions and principal analyses were specified before analysis, and results were retained irrespective of whether they supported the original expectations. Questionnaire invitations were distributed by an independent platform administrator, not by either author, and responses were collected through a secure link that did not display identifying information to the authors. The identifier key was held by that administrator and was not available to the authors during analysis. Participants were told in the information sheet that the research was being conducted by the platform's founder. Parents received the information sheet, consent form and parent questionnaire first; the student questionnaire was released only after parental consent was recorded. Two reminders were sent at seven and 14 days.

4. Results

4.1 Are quality and fit distinct, and where does each vary?

Table 2 reports descriptive statistics, reliabilities and correlations. Quality was rated slightly higher than fit ($M = 3.98$ versus 3.78) and the two were essentially unrelated ($r = .12$, ns), consistent with the .15 factor correlation and the loading pattern in Section 3.5. The same held on the parent side for value and responsiveness ($r = .03$). H1 is supported: quality and fit are not two measurements of one evaluation.

Table 2. Descriptive statistics, reliabilities and correlations among study constructs (matched sample, $n = 134$)

Construct	n (all)	M	SD	Rel.	1	2	3	4	5	6
1. Tutor-student fit	178	3.78	0.67	.747						
2. Tutor quality	178	3.98	0.66	.788	.12					
3. Student satisfaction	178	3.93	0.75	.656 (a)	.52**	.24**				
4. Perceived .	163	4.24	0.74	.819	-.01	.25**	.19*			
5. Platform responsiveness	163	4.21	0.67	.750	.06	.07	.19*	.03		
6. Parent satisfaction	163	4.18	0.66	.583 (a)	.12	.29**	.26**	.47**	.30**	
7. Repeat booking (0/1)	220	.577			.20*	.08	.25**	.07	.12	.24**

Note. Means and SDs on all available cases; correlations on the 134 matched relationships. Rel. = Cronbach's alpha, or (a) Spearman-Brown for two-item scales. Correlations with repeat booking are point-biserial. * $p < .05$, ** $p < .01$.

The stronger test concerns where each attribute varies. In linear random-intercept models with the 24 initial tutors as the grouping factor, perceived quality showed substantial

clustering by tutor ($ICC = .31$), whereas perceived fit showed essentially none ($ICC < .01$), as did student satisfaction. Students taught by the same tutor therefore agreed appreciably about that tutor's quality while showing no detectable agreement about whether the tutor suited them. This is the direction H3 predicts and it is difficult to reconcile with a single underlying evaluation, since one evaluation cannot have a substantial tutor-level component and none at all at the same time. Two limits on the reading should be stated. An ICC of .31 means that 31% of the variance in perceived quality lies between tutors and 69% within them, so quality has a substantial tutor-level component rather than being wholly an attribute of the tutor. And an ICC below .01 shows only that perceived fit does not cluster by tutor; the remaining variance will contain genuine pair-level fit together with student-specific response tendencies and measurement error, which these data cannot separate. What the decomposition supports is that the two constructs are distributed differently across tutors, which is what distinguishing them requires; it does not establish that either is purely one thing. The operational implication follows from the difference rather than from any claim of purity: selection acts on what varies between tutors, and for perceived fit almost nothing does.

Key takeaway. Quality and fit are not two names for one judgement: they are uncorrelated, load separately, and are distributed differently across tutors — quality with a substantial tutor-level component, perceived fit with essentially none.

4.2 Student and parent satisfaction: two dimensions on each side

Table 3 presents the two satisfaction regressions. The student model explained 33.6% of the variance (adjusted $R^2 = .297$; $F(10, 167) = 8.46$, $p < .001$). Fit had the larger association ($b = .562$, $SE = .063$, 95% CI [.437, .686], $\beta = .51$, $p < .001$) and quality retained an independent association ($b = .203$, $SE = .077$, 95% CI [.053, .354], $\beta = .18$, $p = .008$; Holm-adjusted $p = .008$). H2 is supported: neither attribute is redundant given the other. Initial response time had a small negative association ($b = -.013$ per hour, $p = .046$); trial-session status, year group and subject were not significant.

The parent model explained 34.9% of the variance (adjusted $R^2 = .306$; $F(10, 152) = 8.13$, $p < .001$). Perceived value ($b = .438$, $SE = .080$, 95% CI [.282, .594], $\beta = .49$) and platform responsiveness ($b = .311$, $SE = .078$, 95% CI [.158, .465], $\beta = .31$) were both significant after Holm adjustment (both $p < .001$), while the session fee actually paid added no independent association ($b = -.001$, $p = .907$), nor did objective response time ($p = .631$). H4 is supported. Note that this concerns parent satisfaction: the fee added nothing once perceived value and responsiveness were in the model. No test of the relation between fee and perceived value itself is reported here, and none should be inferred.

Both satisfaction models are same-source and cross-sectional, so their coefficients should be read as structural evidence that the constructs separate, not as estimates of effect size.

Table 3. OLS regression models of student satisfaction and parent satisfaction (HC3 standard errors)

Predictor	Student satisfaction (n = 178) b (SE)	95% CI	p	Parent satisfaction (n = 163) b (SE)	95% CI	p
Tutor-student fit	.562 (.063)	[.437, .686]	< .001			
Tutor quality	.203 (.077)	[.053, .354]	.008			
Perceived value				.438 (.080)	[.282, .594]	< .001
Platform responsiveness				.311 (.078)	[.158, .465]	< .001
Session fee (local units)				-.001 (.011)	[-.023, .020]	.907
Initial response time (hours)	-.013 (.007)	[-.026, .000]	.046	.004 (.009)	[-.013, .021]	.631
Trial session (1 = yes)	.021 (.101)	[-.176, .218]	.834			
Year 11 (ref. Year 10)	.043 (.095)	[-.143, .229]	.652	-.146 (.092)	[-.326, .034]	.112
Physics (ref. Mathematics)	.075 (.127)	[-.175, .324]	.557	-.147 (.141)	[-.423, .129]	.297
Chemistry	.294 (.172)	[-.042, .630]	.087	-.182 (.133)	[-.443, .079]	.171
Biology	.103 (.158)	[-.207, .413]	.513	-.207 (.151)	[-.504, .089]	.170
English	.127 (.150)	[-.167, .421]	.397	.006 (.151)	[-.289, .302]	.966
Economics	.085 (.215)	[-.337, .507]	.693	-.081 (.249)	[-.569, .406]	.744
Constant	.947 (.365)		.009	1.195 (.542)		.028
R ² / adjusted R ²	.336 / .297			.349 / .306		
F (df)	8.46 (10, 167)		< .001	8.13 (10, 152)		< .001

Note. Unstandardised coefficients. Holm-adjusted p-values for the hypothesis terms: fit < .001, quality .008, value < .001, responsiveness < .001.

Key takeaway. The same two-dimensional structure appears on both sides of the transaction: fit and quality for the learner; value and responsiveness for the payer, with the fee adding no independent association with parent satisfaction once value is accounted for.

4.3 Behavioural evidence: continuation

The 30-day repeat-booking rate was 57.7% (127 of 220) and the 60-day activity rate 32.3%; in the matched sample, 64.2% and 37.3%. Table 1 shows that families who booked again had slightly younger students (15.35 versus 15.62 years, $p = .014$), paid lower first-session fees (18.87 versus 20.16 units, $p = .037$) and were less likely to have had an early tutor change (7.1% versus 21.5%, $p = .003$). Where the original tutor was retained, 61.8% of relationships produced a repeat booking; where an early change occurred, 31.0%.

Table 4 presents the nested logistic models on the matched sample. Adding student satisfaction to the operational model raised McFadden's pseudo- R^2 from .058 to .097 and improved fit (LR $\chi^2(1) = 6.75$, $p = .009$), with an odds ratio of 1.96 (95% CI [1.16, 3.30], $p = .012$). Adding parent satisfaction instead raised it to .084 (LR $\chi^2(1) = 4.54$, $p = .033$; OR = 1.88, 95% CI [1.04, 3.37], $p = .036$). Student and parent satisfaction each showed an unadjusted association with repeat booking when entered separately, but neither individual coefficient remained significant after Holm adjustment across the four hypothesis-bearing terms in the joint model (student $p = .150$, parent $p = .368$). The pattern is consistent with H5 but does not provide multiplicity-adjusted confirmation of it, and since this paper declares Holm adjustment part of its confirmatory inference, H5 is not treated as confirmed.

On RQ1, objective initial response time was not statistically associated with continuation in any specification (joint model OR = 0.96, 95% CI [0.89, 1.03], $p = .247$). This is a non-significant result and not a demonstration of no association: the confidence interval still admits a reduction of about 11% in the odds of continuation per additional hour, and no equivalence test against a pre-specified negligible effect was conducted. Speed of first reply is a service metric platforms monitor closely, and these data give no evidence that it relates to whether the relationship lasted, which is a weaker statement than showing that it does not.

H6 received support that varies with the sample. Early tutor reassignment was associated with lower repeat booking in the full sample of 220 (Appendix C: OR = 0.29, 95% CI [0.12, 0.70], $p = .006$) and under multiple imputation across all 220 cases (OR = 0.24, $p = .002$), but the matched complete-case estimate was in the same direction and imprecise

(operational model OR = 0.62, 95% CI [0.20, 1.88], $p = .397$; joint model OR = 0.48, 95% CI [0.14, 1.72], $p = .259$), where only 17 of the 29 reassignments are observed. Because reassignment and repeat booking fall inside the same 30-day window, these estimates cannot establish temporal or causal direction. The temporally ordered 60-day analysis, in which the reassignment necessarily precedes the outcome, also showed lower odds of continued activity (OR = 0.31, 95% CI [0.10, 0.95], $p = .041$). The fair summary is that reassignment is a repeatedly observed operational signal of a relationship at risk, not that it has been shown to reduce continuation.

Table 4. Nested logistic regression models of 30-day repeat booking, matched sample (n = 134, 86 events; odds ratios with 95% CI, HC1 standard errors)

Predictor	Model 1 Operational	Model 2 + Student satisfaction	Model 3 + Parent satisfaction	Model 4 Both	Model 4b Both, three-item scales
Early tutor change	0.62 [0.20, 1.88]	0.49 [0.15, 1.57]	0.57 [0.17, 2.00]	0.48 [0.14, 1.72]	0.43 [0.12, 1.56]
Initial response time (hours)	0.95 [0.88, 1.02]	0.95 [0.88, 1.02]	0.96 [0.89, 1.03]	0.96 [0.89, 1.03]	0.97 [0.89, 1.04]
Days to first lesson	0.96 [0.79, 1.18]	0.97 [0.79, 1.19]	0.96 [0.78, 1.17]	0.96 [0.78, 1.19]	0.97 [0.79, 1.20]
Session fee (local units)	0.91 [0.83, 0.99]*	0.93 [0.85, 1.01]	0.93 [0.85, 1.01]	0.94 [0.86, 1.02]	0.94 [0.86, 1.03]
Trial session	0.96 [0.45, 2.08]	0.97 [0.44, 2.16]	0.86 [0.39, 1.89]	0.89 [0.40, 2.00]	0.85 [0.38, 1.92]
Tutor experience (months)	1.00 [0.97, 1.02]	1.00 [0.97, 1.02]	1.00 [0.97, 1.02]	1.00 [0.97, 1.02]	1.00 [0.97, 1.02]
Student satisfaction		1.96 [1.16, 3.30]*		1.78 [1.03, 3.07]*	1.98 [1.10, 3.57]*
Parent satisfaction			1.88 [1.04, 3.37]*	1.62 [0.88, 2.99]	1.99 [1.00, 3.96] ($p = .050$)
McFadden pseudo- R^2	.058	.097	.084	.111	.135
-2 log likelihood	164.65	157.90	160.11	155.46	151.28
LR test vs. Model 1, χ^2 (df), p		6.75 (1), .009	4.54 (1), .033	9.19 (2), .010	
LR test vs. Model 2 / Model 3				2.44 (1), .119 / 4.65 (1), .031	

*Note. Models 1 to 4 use the two-item satisfaction scales; Model 4b repeats Model 4 with the three-item blocks that include the intention items. Subject and year-group controls are excluded from the primary models to preserve events per variable (Section 3.7); a version including them is reported in Appendix D. * $p < .05$.*

Key takeaway. Satisfaction on either side is associated concurrently with a further booking, though not after multiplicity adjustment; early reassignment marks relationships at risk; and the response speed platforms optimise shows no association once a first lesson has happened, on evidence that cannot rule one out.

4.4 Exploratory: whose evaluation carries more information

In the model containing both measures, student satisfaction retained a conventionally significant association with repeat booking (OR = 1.78, 95% CI [1.03, 3.07], $p = .038$) while parent satisfaction did not (OR = 1.62, 95% CI [0.88, 2.99], $p = .123$). Likelihood-ratio tests showed the same asymmetry: adding student satisfaction to the model already containing parent satisfaction improved fit (LR $\chi^2(1) = 4.65$, $p = .031$), whereas the reverse addition did not (LR $\chi^2(1) = 2.44$, $p = .119$). The pattern recurred under multiple imputation (student $p = .016$, parent $p = .425$) and with tutor-clustered standard errors (student $p = .047$, parent $p = .134$).

The asymmetry is consistent across specifications and should still not be read as a demonstration. Neither coefficient survives Holm adjustment; the predictors are correlated and the sample small for separating them; and the parent scale is less reliably measured, which alone would attenuate its coefficient. A difference in measurement precision and a difference in substance are not distinguishable here. The finding is recorded as a direction worth testing on a design built for it, with equally reliable instruments on both sides.

Key takeaway. The student-side asymmetry recurs across every specification and survives none of them formally. It is a direction for a better-instrumented study, not a result.

4.5 Robustness

Seven sensitivity checks are reported in full in Appendix D: three-item satisfaction scales; the intention items entered separately; the 60-day outcome; multiple imputation over all 220 cases; subject and year-group controls; tutor-clustered standard errors and a random-intercept model; and student age. Signs and approximate magnitudes were stable throughout. Two results deserve mention. When the continuation-intention items were entered alongside the evaluative satisfaction scales, the satisfaction coefficients fell to

null (student OR = 1.01, $p = .988$; parent OR = 1.13, $p = .766$) while intention retained marginal associations, compatible with intention standing between satisfaction and behaviour but not shown to be a sequence here. And the random-intercept model that produced the variance decomposition returned fixed-effect odds ratios within .02 of the single-level estimates.

4.6 Open responses in relation to the quantitative findings

Optional open responses were received from 142 students and 128 parents and coded into five categories, with an independent reviewer coding a random 20% (raw agreement 88.9%, Cohen's $\kappa = .85$); distributions are in Appendix B. They are reported here against the quantitative results they bear on rather than as a separate strand.

Against Section 4.1, the open responses show the quality–fit separation in the words families themselves used. Students who did not continue described mismatch rather than incompetence — one wrote that the tutor “knew the subject really well but the lessons moved too fast for me,” which is a high-quality, low-fit description that a composite instructor-quality scale would record as a moderate rating and lose. Students who did continue described fit directly, as in “patient when I got things wrong instead of just moving to the next question.” Tutor-related reasons were the largest category for students on both sides of the outcome (65.2% of continuers, 45.3% of non-continuers), so what separated the groups was not whether the tutor mattered but what about the tutor was being described.

Against Section 4.2, parents distributed their reasons far more evenly across tutor (42.0% and 27.7%), platform (21.0% and 23.4%) and price or value (19.8% and 25.5%) than students did. That is the qualitative counterpart of the parent regression, in which value and responsiveness contributed separately: parents were evaluating a bundle, students a relationship.

Against Section 4.3, roughly a fifth of students and a sixth of parents without a repeat booking cited the end of the examination cycle or reduced need rather than any dissatisfaction. Non-continuation is therefore not a clean measure of a failed relationship, which is one reason the behavioural associations in Table 4 are modest in size and why the age association discussed in Section 5.3 is plausible.

5. Discussion

5.1 Quality is not fit

The principal finding is that perceived quality and perceived fit behave as different attributes. They were close to uncorrelated, separated cleanly in the item structure, and each carried information about student satisfaction that the other did not. Most importantly they were distributed differently: roughly a third of the variance in perceived quality lay between tutors, while perceived fit showed essentially no between-tutor variance at all. Perceived quality therefore has a substantial tutor-level component; perceived fit has none that these data can detect, and varies within a tutor's own caseload.

This matters because the online-learning satisfaction literature has largely measured instructor quality as a single composite (Al-Fraihat et al., 2020; Khan et al., 2023; Martin & Bolliger, 2022). A composite of a tutor-level and a pair-level attribute is still associated with satisfaction, which is why the practice has gone unchallenged, but it conflates two things a platform acts on differently and confounds the interpretation of instructor effects. The result gives empirical content in an educational setting to the vertical–horizontal distinction Du and Lei (2022) develop for matching platforms, and is consistent with evidence that tutors differ systematically in instructional behaviour across caseloads (Wang et al., 2025a, 2025b) while the experience of a lesson remains specific to the pairing.

5.2 Fee is not value

The parent side reproduced the structure. Perceived value and platform responsiveness were unrelated to each other and each was independently associated with parent satisfaction, while the session fee added no independent association once value was in the model. Families paying similar amounts reported different judgements of worth, as Zeithaml's (1988) formulation implies they might. Both parties therefore evaluate tutoring on two independent dimensions rather than one, and in both cases the dimension platforms find easiest to record — a tutor rating, a price — is not the one that carries the association.

5.3 What the behavioural data do and do not show

The most serious constraint on this section is temporal. The design cannot separate antecedents of continuation from correlates of it. Questionnaires were completed within 15 days of the first lesson and the outcome window is 30 days, so for an unknown share of families the sequence may run the other way: a family that has already booked again may then report higher satisfaction, as post-decision consistency would predict. The analytical file does not contain exact questionnaire completion dates, so the proportion of

such cases cannot be estimated from these data; that figure would require completion timestamps that were not retained. This is why every association in this section is described as concurrent rather than predictive, and why the mediation pathway from fit through satisfaction and intention to booking — which sensitivity check 2 makes tempting — is not tested here. Satisfaction was associated with a further paid booking from either side, but the association did not survive multiplicity adjustment in the joint model, so it is consistent with expectation-disconfirmation accounts (Bhattacharjee, 2001; Oliver, 1980) rather than a confirmation of them. Early reassignment appeared as a marker of relationships at risk across the full-sample, imputed and 60-day analyses, though not in the matched subsample; it is best described as a repeatedly observed operational signal. Objective response time, which platforms monitor and optimise, showed no association with continuation among families who had already had a lesson; on these data that is an absence of evidence rather than evidence of absence. Whether the student's or the parent's evaluation carries more unique information remains open.

One administrative association deserves more attention than it initially received. In the full operational model, each additional year of student age was associated with about 30% lower odds of a repeat booking (OR = 0.70, 95% CI [0.49, 0.98], $p = .039$), and Year 11 students were somewhat less likely to continue than Year 10 students. The straightforward reading is not that older students are harder to satisfy but that they are further through a two-year examination cycle: a Year 11 student in the second half of the IGCSE course has a fixed and approaching endpoint, after which tutoring in that subject stops whatever the family thinks of it. The open responses support this, with about a fifth of students who did not continue naming the end of the examination cycle or reduced need. This matters for how the outcome should be read. In a curriculum with a terminal examination, non-continuation is partly a calendar event, and any study using continued purchase as an outcome in such a setting should model or at least report position in the examination cycle. Age was not pre-specified here and is reported as an observation rather than a test.

5.4 Implications for platforms

If quality carries a tutor-level component and fit does not, screening and matching are not substitutes, and a platform investing only in the average quality of its tutor pool is acting on the part of the variation that selection can reach while leaving the larger part untouched. Fit is observable only once a pairing exists, so it must be measured early and at the level of the pair, which a tutor rating aggregated across students cannot do. And because reassignment marks relationships already in difficulty, the operational question is whether mismatch can be detected before that point. A short pair-level check after the

first or second lesson, asking about pace and suitability rather than the tutor's competence, is the direct implication of these results and is testable.

5.5 Limitations

The design is observational and single-site — one platform, one qualification, one age band — so generalisation beyond comparable online supplementary tutoring is not warranted. The satisfaction models are same-source and cross-sectional and cannot support causal claims. Questionnaire completion and the booking outcome are not strictly ordered, so a repeat booking may sometimes have preceded the questionnaire. Response was associated with the outcome, tilting the questionnaire sample towards families who continued. Both satisfaction scales are two-item, and the parent scale is weak (Spearman–Brown .583), which limits the student–parent comparison specifically. Tutor type was not recorded, so peer and professional tutors could not be compared, and 24 tutors is few for cluster-based inference. The variance decomposition that carries the main claim is least affected by these limitations, since it concerns the distribution of one construct across tutors rather than a relationship between self-reported variables, but it too rests on 24 tutors and should be replicated.

Two contextual features of the setting deserve separate mention, because they bound the findings in ways a larger sample would not fix. First, the platform is a single provider working in one qualification, IGCSE, with a terminal examination; the findings should not be assumed to carry to primary tutoring, university tutoring, test-preparation services, large commercial providers or AI-assisted tutoring platforms, each of which differs in how continuation arises and what it means. Second, the tutor pool mixes peer tutors, typically students who recently sat the same qualification, with adult or professional tutors. That mix is unusual and is plausibly relevant to the central result: peer tutors may vary less in the instructional behaviours that constitute quality while varying more in the pace and manner that constitute fit, or the reverse. Tutor type was not recorded in the administrative data, so it could be neither controlled nor tested as a moderator, and the reported variance decomposition is an average across a heterogeneous pool. Collecting tutor type is the single cheapest improvement a replication could make.

6. Conclusion

In one online IGCSE tutoring platform, how good a tutor is and how well that tutor suits a particular student behaved as separable attributes. Perceived quality clustered substantially by tutor; perceived fit did not, varying instead within a tutor's own caseload; and each was independently associated with student satisfaction. The parent side separated the same way, into perceived value and platform responsiveness, with the

session fee adding no independent association with parent satisfaction once value was accounted for. These findings are associational and from a single setting, but they carry a straightforward implication for platforms that recruit and rank tutors as their main quality lever: selecting better tutors and matching them better are different problems, because the two attributes are not distributed at the same level.

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Authors' contributions

KG conceived the study, designed the analysis, conducted the statistical analysis and wrote the manuscript. AA provided the study setting and the administrative records, contributed to the design of the questionnaires, and reviewed and approved the manuscript. Both authors read and approved the final version.

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Competing interests

The second author, Arhaan Anand, is the founder of the online tutoring platform from which the data were obtained. This relationship provided access to operational records and also constituted a potential conflict of interest; the safeguards adopted are set out in Section 3.8. The first author, Karan Gupta, holds professional roles in education and career counselling and has no financial or other interest in the tutoring platform. No participants were recruited through the first author's professional practice.

Informed consent

Obtained. All participating students were under 18; written parental or guardian consent was obtained together with student assent before participation.

Ethics approval

Granted by the Zigma Six Ethics Review Board, India, on 15 April 2026 (Approval No. SS/2026/002-23; OHRP IRB00015192, Federalwide Assurance FWA00036459).

Data availability statement

The pseudonymised data supporting the findings are available from the corresponding author on reasonable request, subject to the terms of participant consent and ethical approval. Because the participants are minors and the records are commercially derived, the data are not deposited in a public repository.

Declaration of generative AI use

The authors used Claude (Anthropic) and ChatGPT (OpenAI) during manuscript preparation for restructuring, language editing and critique of drafts. The authors designed the study, conducted all analyses, verified all outputs and take full responsibility for the content of the published article.

Provenance and peer review

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Appendices

Supplementary material for “Tutor quality is not tutor–student fit: Distinct sources of satisfaction in online secondary-school tutoring”.

Appendix A. Questionnaire items by construct

Table A1. Questionnaire items by construct

Construct (respondent)	Item
Tutor-student fit (student)	FIT1 My tutor's teaching style suited the way I learn. FIT2 My tutor understood the areas where I needed the most help. FIT3 The pace of the tutoring sessions was right for me. FIT4 Overall, the tutor was a good match for my learning needs.
Tutor quality (student)	TQ1 My tutor explained difficult ideas clearly. TQ2 My tutor came prepared and the sessions felt organised. TQ3 My tutor gave me useful feedback on what I needed to improve. TQ4 My tutor changed the explanation or approach when I did not understand.
Student satisfaction (student)	SS1 Overall, I was satisfied with the tutoring I received. SS2 The tutoring experience was as good as or better than I expected.
Student continuation intention (student; not in scale)	SS3 If I needed help in this subject again, I would be happy to continue with this tutor.
Perceived value (parent)	PV1 The fee was reasonable for the quality of tutoring my child received. PV2 The tutoring provided good value for the amount we paid. PV3 The fees and any additional charges were clear before we committed.
Platform responsiveness (parent)	PR1 It was easy to arrange the first tutoring session. PR2 The platform responded quickly when I had a question or problem. PR3 Scheduling changes or tutor-related issues were handled effectively.
Parent satisfaction (parent)	PS1 Overall, I was satisfied with the tutoring service. PS2 The service met the expectations I had when I signed my child up.
Parent continuation intention (parent; not in scale)	PS3 I would use the platform again if my child needed tutoring.

Note. Five-point agreement scale for all items (1 = strongly disagree, 5 = strongly agree). The two continuation-intention items (SS3, PS3) are excluded from the primary satisfaction scales and analysed separately (Appendix D, check 2).

Appendix B. Open-response distributions

Table B1. Main reason given by students, by 30-day repeat-booking status (n = 142)

Main reason identified	Students with repeat booking (n = 89)	Students without repeat booking (n = 53)
Tutor-related	58 (65.2%)	24 (45.3%)
Platform-related	7 (7.9%)	8 (15.1%)

Price or value	3 (3.4%)	4 (7.5%)
Exam cycle or tutoring no longer required	14 (15.7%)	12 (22.6%)
Other	7 (7.9%)	5 (9.4%)

Table B2. Main reason given by parents, by 30-day repeat-booking status (n = 128)

Main reason identified	Families with repeat booking (n = 81)	Families without repeat booking (n = 47)
Tutor-related	34 (42.0%)	13 (27.7%)
Platform-related	17 (21.0%)	11 (23.4%)
Price or value	16 (19.8%)	12 (25.5%)
Exam cycle or tutoring no longer required	9 (11.1%)	8 (17.0%)
Other	5 (6.2%)	3 (6.4%)

Note. Each response was assigned to the category representing its principal reason. An independent reviewer coded a random 20% of responses (54 of 270): raw agreement 88.9%, Cohen's $\kappa = .85$. Disagreements were resolved before the remaining responses were classified.

Appendix C. Models on the full available samples

The models reported in Table 4 of the main text are estimated on the 134 matched relationships, so that differences in fit between nested models reflect the variables added rather than a changing sample. Table C1 reports the corresponding models on the largest sample available for each specification. The early tutor-change association is stronger and conventionally significant in the full operational sample, where all 29 reassignments are observed, than in the matched subsample, where only 17 are.

Table C1. Logistic regression models of 30-day repeat booking on the full available samples (odds ratios with 95% CI, HC1 standard errors)

Predictor	Operational (n = 220, 127 events)	Student (n = 178, 109 events)	Parent (n = 163, 101 events)
Early tutor change	0.29 [0.12, 0.70]**	0.36 [0.12, 1.11]	0.42 [0.14, 1.23]
Initial response time (hours)	0.97 [0.92, 1.02]	0.95 [0.90, 1.01]	1.00 [0.93, 1.08]
Days to first lesson	1.03 [0.89, 1.19]	0.92 [0.76, 1.10]	1.09 [0.91, 1.30]
Session fee (local units)	0.94 [0.88, 1.01]	0.94 [0.87, 1.01]	0.94 [0.87, 1.01]
Trial session	1.17 [0.66, 2.07]	1.12 [0.57, 2.22]	0.91 [0.46, 1.82]
Tutor experience (months)	1.00 [0.98, 1.01]	1.00 [0.98, 1.02]	1.00 [0.98, 1.02]
Tutor-student fit		1.46 [0.79, 2.67]	

Tutor quality		1.10 [0.67, 1.82]	
Student satisfaction		1.69 [0.99, 2.90] (p = .056)	
Perceived value			0.84 [0.49, 1.45]
Platform responsiveness			1.13 [0.63, 2.02]
Parent satisfaction			1.66 [0.92, 3.02] (p = .093)
McFadden pseudo-R ²	.050	.105	.054
-2 log likelihood	284.60	212.74	204.76

Note. ** $p < .01$. Each column uses the largest sample for which its predictors are observed.

Appendix D. Sensitivity and robustness checks

1. Three-item satisfaction scales. Re-estimating the joint model with the original three-item satisfaction blocks, which include the continuation-intention items, produced stronger coefficients for both parties (student OR = 1.98, 95% CI [1.10, 3.57], $p = .023$; parent OR = 1.99, 95% CI [1.00, 3.96], $p = .050$; McFadden pseudo-R² = .135). These scales are not used in the primary analyses because the intention items overlap conceptually with the behavioural outcome.

2. Intention items entered separately. The two intention items entered on their own were associated with repeat booking (student OR = 1.88, $p = .012$; parent OR = 2.11, $p = .014$; pseudo-R² = .153). Entered alongside the two-item evaluative satisfaction measures, the satisfaction coefficients fell to null (student OR = 1.01, $p = .988$; parent OR = 1.13, $p = .766$) while the intention items retained marginal associations ($p = .076$ and $.075$). Intention therefore accounted for much of the observed association between evaluative satisfaction and repeat booking. This is compatible with theories placing intention between satisfaction and behaviour, but the data cannot establish that sequence, because questionnaire completion and repeat booking were not strictly ordered for every family.

3. Sixty-day outcome. Substituting activity in days 31 to 60 (50 events) left coefficients in the same direction but imprecise (student OR = 1.64, $p = .100$; parent OR = 1.36, $p = .316$). In the full sample the only significant 60-day operational association was early tutor change (OR = 0.31, 95% CI [0.10, 0.95], $p = .041$).

4. Multiple imputation. Imputing questionnaire scores for all 220 relationships (20 imputations, chained equations with the administrative variables and the repeat-booking outcome as auxiliaries, combined by Rubin's rules) gave student satisfaction OR = 1.80 (95% CI [1.12, 2.91], $p = .016$), parent satisfaction OR = 1.21 (95% CI [0.76, 1.95], $p =$

.425), early tutor change OR = 0.24 (95% CI [0.10, 0.60], $p = .002$) and response time OR = 0.98 ($p = .469$).

5. Subject and year-group controls. Adding these dummies to the joint model left the pattern broadly unchanged (student OR = 1.83, $p = .039$; parent OR = 1.84, $p = .063$), at the cost of events per variable falling to about six.

6. Tutor clustering. With standard errors clustered by the 24 initial tutors, every hypothesis-bearing coefficient kept its sign and similar magnitude: student satisfaction OR = 1.78, 95% CI [1.01, 3.15], $p = .047$; parent satisfaction OR = 1.62, 95% CI [0.86, 3.04], $p = .134$; early tutor change in the full operational model OR = 0.29, 95% CI [0.12, 0.69], $p = .005$; and the fit, quality, value and responsiveness coefficients remained significant at $p \leq .001$. A tutor random-intercept logistic model (variational Bayes) estimated a between-tutor standard deviation of 0.47 on the logit scale in the matched sample (ICC .06) and 0.28 in the full sample (ICC .02), with fixed-effect odds ratios within .02 of the single-level estimates. The linear random-intercept models reported in Section 4.1 of the main text come from the same specification: tutor-student fit and student satisfaction had essentially no between-tutor variance (ICC < .01) whereas tutor quality had ICC = .31.

7. Student age. Age differed between repeat-booking groups in Table 1 but was not pre-specified. Added to the joint model it was not significant (OR = 0.73 per year, $p = .240$) and left the satisfaction coefficients similar (student OR = 1.82, $p = .026$; parent OR = 1.61, $p = .133$). In the full operational model age was associated with lower repeat booking (OR = 0.70 per year, 95% CI [0.49, 0.98], $p = .039$), plausibly because older students were nearer the end of the examination cycle; the tutor-change estimate remained similar (OR = 0.31, $p = .011$).